

Epigenetics

Paper: [Lux, V. \(2018\) Epigenetic Programming Effects of Early Life Stress: A Dual-Activation Hypothesis. Current Genomics 19, 638-652.](#)

Background & Context (Sections 1 & 2)

Why this paper exists

Early life stress (ELS)—such as maternal separation, neglect, prenatal stress, or early sensory threat—is strongly associated with long-term changes in:

- Stress responsivity
- Emotion regulation
- Cognition
- Risk for anxiety, depression, and PTSD

A central biological question is:

How can experiences early in life produce effects that last into adulthood?

The authors argue that traditional models focusing only on hormones or neural activity are insufficient. Instead, they propose that **epigenetic mechanisms** stabilize early experiences into long-term biological changes.

Why would changes in hormone levels or neural activity alone be insufficient to explain effects of early life stress that last for decades?

Core idea introduced in Section 1

Early life stress does not act through a single pathway.

Instead, it **simultaneously activates**:

- **Stress-regulatory systems** (e.g., HPA axis, glucocorticoids)
- **Neural and sensory systems** that process the stressor itself

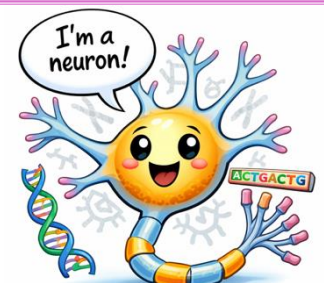
When this occurs during **critical periods of development**, the resulting epigenetic changes can persist long after the stressor is gone. This framework underlies the paper's central model (developed later) known as the **dual-activation hypothesis**.

What “Epigenetics” Means in This Paper (Section 2)

In the nervous system, epigenetic mechanisms operate at multiple functional scales. **Structural epigenetic mechanisms** establish and maintain neuronal identity by stably regulating large gene programs (for example, keeping neuron-specific genes active and non-neuronal genes silent across the lifetime of a neuron). **Synaptic epigenetic mechanisms** act on much shorter timescales, using reversible chromatin modifications to support activity-dependent gene expression during learning, memory formation, and moment-to-moment neural plasticity. A third category—**developmental epigenetic mechanisms**—links early experience to long-term changes in brain function and is the primary focus of this paper.

Structural Epigenetic Mechanisms

These mechanisms maintain **genomic stability and cell identity** over long periods of time. In neurons, they ensure that neuron-specific genes remain active and that genes inappropriate for neurons remain silent. While essential for nervous system function, structural epigenetic mechanisms are **not the main focus** of this paper.





Synaptic Epigenetic Mechanisms

These mechanisms support **short-term, activity-dependent plasticity** in neurons. They involve reversible chromatin modifications that allow neurons to rapidly adjust gene expression in response to synaptic activity during learning and memory, without permanently altering developmental trajectories.

Developmental Epigenetic Mechanisms (*Focus of This Paper*)

These mechanisms operate during **critical or sensitive periods of development** and are relatively stable over time. By altering chromatin accessibility during early development, they can shift long-term neural circuit formation and function, providing a mechanism by which early life experiences become biologically embedded.



Why timing matters

A central claim of the paper is:

The same epigenetic modification has very different consequences depending on when it occurs.

During early development → can permanently alter brain function

After development → more likely to be transient or reversible

This explains why early life stress is uniquely powerful compared to stress experienced later in life.

Systems-level perspective

The authors also introduce **systems heritability**:

Epigenetic changes in one neural system can drive lasting changes in connected systems via neural activity

Epigenetic programming does not need to occur everywhere to have widespread effects

This supports the idea that early life stress can shape:

Stress-regulatory circuits

Sensory systems

Cognitive and emotional networks together

Why might epigenetic changes in one neural system be sufficient to alter behavior, even if other systems are not directly modified?

What to keep in mind as you read later sections

Early experiences can become biologically embedded

Epigenetic mechanisms provide stability to those experiences

When epigenetic changes happen (developmentally) matters

Stress affects **interacting systems**, not just stress hormones

In your group work, you will examine **one specific epigenetic mechanism** and connect it back to this framework.

As you read about specific genes and epigenetic marks, how will you decide whether a change represents short-term regulation or long-term developmental programming?

How this connects to your group activity

In your group, you will examine **one specific epigenetic system** described later in the paper.

As you work, keep asking:

How does this specific mechanism fit into the broader framework described in Sections 1 and 2?

Activity

Your group leader will assign your paper section

Assignment	Paper Sections to Read
1	Section 3.1
2	Section 3.2
3	Section 3.2 (Crf portion only) and Crf-related mechanisms in Section 5.2
4	Section 3.3 and FKBP5 discussion in Section 5.2
5	Section 4 (BDNF paragraphs only) and BDNF-related sensory development in Section 6
6	Section 5.1

What sections of the paper has your group been assigned to explore?

Write the section name(s) as the title of your ‘poster’

Create a poster section for ‘words I didn’t know’. As you read, **look those words up** and define them on your poster

What is the biological system being discussed? (e.g. hippocampus, Para ventricular nucleus, sensory cortex)

Look it up: Where is this system and what are a few things this system does in the human body (what does it regulate, control, influence?)

On your poster, create a 'system's affected' section.

Sketch a diagram/illustration of the system and its location in the body.

Visually represent a few things this system does.

What environmental input (early life stressor) is affecting the system (e.g. maternal separation, cortisol levels, prenatal stress, sensory stimuli)?

On your poster, represent this input in a way that makes sense to you

What epigenetic modification is described?

On your poster, create an 'epigenetic modifications' section.

You will be using binder clips and/or push pins to represent epigenetic modifications.



Which of the options will represent which epigenetic modification(s)?

On your poster, draw and label a representation of the modification(s).

Where in the genome does this modification occur?

In your 'epigenetic modifications' section create a visual representation of the location of the epigenetic modification(s). Label it accordingly

How does this modification affect gene expression?

Next to the epigenetic modification(s) & their location(s) indicate if the modification increases or decreases gene expression.

What does this do to the body in which this modification takes place long term?

Build the Model

DNA is always wrapped around histones. 'Open' and 'closed' chromatin describe how accessible the DNA is to other proteins, not whether it is wrapped or unwrapped.

Object	Represents
Plastic 'string'	DNA
8 styrofoam balls	Histone octamer
Pipe cleaners	Histone tails

Note: your group may not have all of the below to represent depending on which section you were assigned

Object	Represents
	Methylation
	Methylation
	Acetylation
	HDACs / repressors
	Transcription factors

Create a model of your epigenetic target in an environmental-input neutral state (no stressor in early life - page 3 if you don't remember the environmental input).

Change the model to represent how it changes epigenetically in response to the environmental input

What physically changed about your model?

How does this change gene accessibility?

Which part of this system would you target if you wanted to *reverse* the effects of early life stress—and why?

What is one different epigenetic change that could accomplish the same change to gene accessibility?

Model the other epigenetic change that could accomplish the same change to gene accessibility.

Demo

The class will move to each group's workstation to view a demo that demonstrates:

“Here's what the DNA/chromatin looks like before the environmental input (early life stressor)”

“Here's what the DNA/chromatin looks like when affected by early life stress”